

INTERFACE CONTROL DOCUMENT

Warthrop Systems LLC — Payload Integration ICD

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1. PURPOSE

This Interface Control Document defines the hardware and software interface requirements for the Warthrop Model WX-7 Sensor Payload as it integrates with the government-provided bus structure. This document is proprietary to Warthrop Systems LLC and shall not be distributed to other contractors or third parties without written authorization.

2. BUS INTERFACE SPECIFICATION

2.1 Data Bus Protocol

The WX-7 payload communicates over a MIL-STD-1553B dual-redundant serial data bus. All communication shall conform to MIL-STD-1553B Rev B specifications.

2.2 Message Rate

The WX-7 transmits telemetry at a nominal rate of 100Hz. The bus controller shall be capable of accommodating 100 message transactions per second per RT address.

2.3 Error Handling

All data words transmitted by the WX-7 shall include a CRC-16 checksum appended to the data frame. The receiving controller must validate the CRC-16 checksum before processing any message. Messages failing CRC-16 validation shall be discarded and flagged in the fault log.

2.4 Word Length

All data words are 16-bit fixed length in accordance with MIL-STD-1553B word structure.

2.5 Voltage Tolerance

Bus voltage levels shall be maintained within +/-2% of nominal. The WX-7 payload will generate a fault condition if bus voltage deviates beyond this tolerance.

3. CONNECTOR SPECIFICATION

All electrical connections between the WX-7 payload and the bus structure shall use MIL-DTL-38999 Series III circular connectors. Pin assignments are defined in Appendix A.

4. ENCRYPTION

All data transmitted over the 1553B bus shall be encrypted using AES-256 in CBC mode. Encryption keys shall be loaded prior to mission initialization via the key loading interface defined in Section 7.

5. ENVIRONMENTAL REQUIREMENTS

The WX-7 payload is rated for operating temperatures of -40C to +85C. The bus interface hardware must maintain nominal operation across this temperature range.

6. NOTES

Contact Warthrop Systems engineering at integration@warthrop-sys.com for clarification on any interface requirement defined in this document.